

QTPTC1 Module 2

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The Deutsch-Jozsa algorithm

This lab showed how quantum circuits can use superposition and phase kickback to learn global properties of black box functions more efficiently than deterministic classical methods. The starting point was a two qubit oracle U_f that computes $f(x)$ into a target bit through $U_f|y\rangle|x\rangle = |y \oplus f(x)\rangle|x\rangle$. Initializing the input in the superposition $|+\rangle$ and applying U_f prepares a state where both $f(0)$ and $f(1)$ are encoded at once. Measuring both qubits just returns a single classical pair $(x, f(x))$, collapsing the state. Repeating the experiment showed that this simple use of superposition does not improve query complexity over a classical strategy, since multiple shots are still needed to see both inputs and decide whether f is balanced or constant.

Deutsch's algorithm then introduced an oracle acting on one input qubit and one ancilla qubit, with the ancilla prepared in $|1\rangle$, rotated to $|-\rangle$ by a Hadamard. A Hadamard on the input qubit created an equal superposition of $|0\rangle$ and $|1\rangle$, the oracle applied the map U_f , and a final Hadamard on the input qubit produced an output that depended only on whether f was constant or balanced. Running the circuit with each of the four oracles on a simulator gave a single measurement of the input qubit that was always $|0\rangle$ for constant oracles and always $|1\rangle$ for balanced ones. This confirmed that the algorithm computes the parity $f(0) \oplus f(1)$ in exactly one query, while a deterministic classical method needs two evaluations.

The same structure was generalized to the Deutsch-Jozsa algorithm with n input qubits and one ancilla. A random oracle U_f was built that was guaranteed to be either constant or balanced on the 2^n possible inputs. After preparing the ancilla in the state $|1\rangle$ and applying a Hadamard gate to each qubit (input and ancilla), the oracle applied f across all inputs. A second layer of Hadamards applied only to the input qubits mapped the phase information onto a bit string. If the resulting bit string contained at least one 1, the function was balanced, while an outcome of all 0's indicated a constant function. This was then tested on quantum hardware with three input qubits. This result was obtained with a single oracle query, compared to $2^{n-1} + 1$ queries in the worst case for any deterministic classical algorithm.

Finally, the Bernstein-Vazirani problem was implemented using the same Hadamard-oracle-Hadamard pattern with a different oracle. The function was of the form $f(x) = s \cdot x \pmod{2}$, where s is an unknown n bit string. An oracle was constructed by applying a CNOT from each input qubit x_i with $s_i = 1$ to the ancilla. With the ancilla prepared in $|-\rangle$, each basis state $|x\rangle$ then acquired a phase $(-1)^{s \cdot x}$ through phase kickback. After the final Hadamards on the inputs, the measurement result on the input register returned the hidden string s directly. This was demonstrated for a four bit secret string, where a single shot produced the correct output bit string on the input qubits. In this oracle model a classical algorithm must query f on n different inputs to recover s , while the Bernstein-Vazirani circuit recovers all bits in one query.

This lab showed that quantum algorithms need to combine superposition with phase to outperform classical methods, and that simply putting inputs into superposition without using phase information offers no advantage. Preparing the ancilla in $|-\rangle$ makes the oracle write information about $f(x)$ into phases on the input register. The final Hadamards convert these phases into deterministic measurement patterns through interference. Phase kickback lets quantum circuits extract global properties of functions, such as parity or a hidden string, using fewer queries than any deterministic classical algorithm.

Quantum Teleportation

This lab demonstrated that an unknown qubit state can be transferred from Alice to Bob using entangled qubits and two classical bits. A three qubit circuit was built containing the input qubit to be teleported, Alice's share of an entangled pair, and Bob's qubit. First, a Bell pair between Alice's and Bob's qubits was prepared with a Hadamard on A , followed by a CNOT with A as control and B as target. A random

single qubit state $|\psi\rangle$ was prepared on the input qubit as the unknown state to be teleported. 'Alice' then performed a CNOT from Q to A and a Hadamard on Q , and then measured A and Q in the computational basis, which implements a Bell basis measurement on (A, Q) . If Alice's result on A was 1, an X gate was applied to B , and if the result on Q was 1, a Z gate was applied, leaving Bob with the unknown state $|\psi\rangle$.

To verify that the protocol reproduced $|\psi\rangle$, 'Bob' applied the inverse of Alice's preparation unitary to his qubit and then measured it. If teleportation was successful this should return $|0\rangle$. Running the circuit on quantum hardware, teleportation was successful in the majority of cases, up to hardware noise effects. The protocol did not clone the state, since the measurements on Q and A collapsed those qubits to definite computational basis states and destroyed the original Bell pair.

In the second experiment the same teleportation logic was tested when Alice and Bob were separated across the processor. An initial Bell pair was created between Alice's qubit and Bob's qubit and then moved apart using a sequence of SWAP gates. After this separation step, the same teleportation procedure was followed as before. The deeper circuit showed more hardware noise, but teleportation remained effective.

The experiments showed how Bell pairs, Bell basis measurements, and classically controlled corrections work together to move quantum information without moving the physical qubit, while respecting the no cloning theorem and the light speed limit imposed by the required classical communication.